

Achyuth Kumar Silasagaram

San Jose, CA | achyuth.uf237@gmail.com | 216-860-7167 | linkedin.com/achyuthkumar | Portfolio

Experience

Software Engineer, Kaiser Permanente

Jul 2024 – Present

- Engineer a real-time patient data streaming pipeline using **Kafka** and **AWS Lambda** to process clinical events, reducing data latency from batch cycles to under **20 seconds** for critical patient alerts.
- Develop and deploy a HIPAA-compliant **RAG** system using **LangChain**, **OpenAI** embeddings, and a **PostgreSQL** vector store to augment clinician decision support, improving diagnostic accuracy by **15%**.
- Architect a secure microservices platform on **EKS** using **FastAPI** and **Terraform**, implementing **OAuth2** with **Okta** to centralize patient data access and achieve **40%** faster API response times.
- Orchestrate infrastructure as code with **Terraform** to manage **AWS resources** (S3 data lakes, DynamoDB tables), automating compliance checks and reducing manual provisioning efforts by **90%**.

Software Engineer Intern, Kaiser Permanente

Jan 2024 – Jun 2024

- Developed and integrated responsive UI components for a patient portal using **React** and **TypeScript**, enhancing usability and increasing patient engagement metrics by **25%**.
- Implemented secure **RESTful** endpoints with **Node.js** and **Express** to serve patient wellness data, applying HIPAA-aware patterns and **JWT** authentication, reducing data fetch times by **300 ms**.
- Wrote comprehensive unit and integration tests for frontend components and APIs using **Jest** and **Playwright**, increasing code coverage from **60%** to **85%**.

Software Engineer, Dell Technologies

Mar 2021 – Jul 2022

- Migrated a monolithic legacy application to a containerized microservices architecture using **Spring Boot** and **Docker** on **Kubernetes**, improving scalability and reducing deployment times by **40%**.
- Optimized high-throughput service performance by implementing **Spring WebFlux** and integrating **Redis** caching, decreasing average API latency from **200 ms** to **50 ms** under load.
- Automated the **CI/CD** pipeline using **Jenkins** and Argo CD to enable zero-downtime blue-green deployments, improving release frequency and developer productivity.

Software Engineer, TatvaSoft

Jan 2020 – Feb 2021

- Built a real-time monitoring dashboard with **React**, **TypeScript**, and **Recharts** consuming **WebSocket** data streams to visualize metrics, helping sustain a **99.9%** uptime SLA.
- Implemented a centralized observability platform using the **Prometheus**, **Grafana**, and **Loki** stack, reducing **MTTR** by **30%** through improved tracing and alerting.
- Established a robust **CI/CD** workflow with **GitHub Actions** and **SonarQube**, integrating end-to-end testing with **Cypress** to maintain over **95%** code coverage.

Projects

E-commerce Payment Integration — RESTful APIs, OAuth2, JWT

- Built secure, RESTful APIs with OAuth2 and JWT for third-party payment processing, aligned with PCI-style security.

Real-Time Data Visualization Dashboard — React, Tailwind CSS, Spring Boot, Kafka

- Engineered a responsive frontend with React and Tailwind CSS for real-time data visualization, improving analytic insights.
- Architected a Spring Boot backend for large-scale real-time data aggregation via Kafka.

Technical Skills

Programming Languages: Python, Java, TypeScript, JavaScript, SQL, C, C++, HTML, CSS.

Frameworks & Libraries: React, Angular, Spring Boot 3, FastAPI, Node.js (Express), REST, GraphQL, OpenAPI, Swagger.

Cloud & DevOps: AWS (Lambda, EKS, API Gateway, IAM), Docker, Kubernetes, Terraform.

Databases & Streaming: PostgreSQL, DynamoDB, Redis, Kafka, Amazon S3.

Developer Tools: Git, GitHub, GitHub Actions, Jenkins, PyTest, JUnit 5, Mockito, Jest, Postman.

Monitoring & Security: Prometheus, Grafana, ELK, Splunk, OAuth2, OIDC, JWT.

AI/ML Technologies: LangChain, LangGraph, FAISS, OpenAI embeddings.

Education

Eastern Illinois University — Master of Science in Computer Technology

Aug 2022 – Jun 2024

Vardhaman College of Engineering — Bachelor of Technology in Computer Science

Aug 2017 – Jul 2021

Coursework — Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Web Technologies, Advanced Computer Architecture, Advanced Operating Systems, Cloud Computing, Artificial Intelligence and Computer Networks.